

Python Projects

— *FizzBuzz* code | Level 3 Data Technician Course, Unit 5 —

```
▶ for n in range(1,101):  
    if n % 3 == 0 and n % 5 == 0:  
        print("FizzBuzz")  
    elif n % 3 == 0:  
        print("Fizz")  
    elif n % 5 == 0:  
        print("Buzz")  
    else:  
        print(n)  
  
... 1  
     2  
     Fizz  
     4  
     Buzz  
     Fizz  
     7  
     8  
     Fizz  
     Buzz  
    11  
    Fizz  
    13  
    14  
    FizzBuzz  
    16  
    17
```



```
for number in range(0,101):  
    number = int(input("Is the number divisible by 3, 5, or both? "))  
    if number % 3 == 0 and number % 5 == 0:  
        print("FizzBuzz")  
    elif number % 3 == 0:  
        print("Fizz")  
    elif number % 5 == 0:  
        print("Buzz")  
    elif number in range (0, 101):  
        print(number)  
    else:  
        print("not valid")
```

```
... Is the number divisible by 3, 5, or both? 101  
not valid  
Is the number divisible by 3, 5, or both? 6  
Fizz  
Is the number divisible by 3, 5, or both? 30  
FizzBuzz  
Is the number divisible by 3, 5, or both? 10  
Buzz  
Is the number divisible by 3, 5, or both? 4  
4  
Is the number divisible by 3, 5, or both? 
```

— *EcoDispose Program* | Level 2 IT and Digital Skills Course, Unit 2 —

```
#the privacy policy
def get_greeting():
    return("PRIVACY POLICY")
message = get_greeting()
print(message)
print("Welcome to EcoDispose, your hub of info on electrical
waste recycling and disposal! :)")
print("\n")
print("We collect the following data for the purpose of
providing tailored recycling recommendations:")
print("-> Your name: This is only used for personalisation
purposes during this session")
print("-> The item you wish to recycle: This helps us guide
you through the appropriate disposal or recycling process")
print("-> The condition of the item: Used to give more
specific advice on whether to donate, sell, or recycle")
print("-> Your location: We use this to provide local council
guidelines for recycling and disposal")
print("-> We do not share or store your personal data with
third parties")
print("-> Your data will only be used within this session and
will not be retained afterward")
print("-> For any questions about data protections, contact us
at support@greentechsolutions.com")
print("\n")
#entering basic information
print("\n")
individ = ("Individual")
org = ("Company")
tower_of_memories = input("Are you an " + (individ) + " or " +
(org) + "? ")
select = input("What is your name? ")
print("Welcome" + (select) + ", " + (tower_of_memories))
print("\n")
#defining the options and their corresponding responses
print("Here are a list of multiple-choice questions for you to
go through on your \"
    \"journey to learn about electrical waste disposal.\")
questions = {
    "Informational Purposes": {
        "Info on WEEE": {
            "What does it mean?": {
                "answer": "WEEE stands for 'Waste Electrical
and Electronic Equipment'. " \
                    "It exists because electronics are
complicated to dispose of safely. " \
                    "Waste Direct provides an in-depth
look to what defines WEEE",
```

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        "wastedirect":
"https://wastedirect.co.uk/blog/blog-what-is-weee-waste-in-uk/"
    },
    "Why is it important?": {
        "answer": "This is one way of saving natural
resources and curbing those nasty " \
                    "climate-change emissions. It also
prevents your devices or " \
                    "appliances being another another
addition to the landfill. " \
                    "Restore Technology and Site Waste
Solutions go into more depth " \
                    "regarding the question",
        "links": {
            "restore":
"https://www.restore.co.uk/technology/why-weee-disposal-is-imp
ortant-for-businesses/",
            "site_waste":
"https://www.sitewastesolutions.co.uk/post/weee-" \

"waste-what-is-it-and-why-is-it-important-to-recycle-it"
        },
    },
    "What is NOT included under WEEE": {
        "answer": "Here, NetRegs goes into detail on
what IS and IS NOT included under WEEE",
        "netregs":
"https://www.netregs.org.uk/environmental-topics/waste/" \

"wEEE-waste-electrical-and-electronic-equipment/equipment-cove
red-by-the-weee-regulations/"
    },
    "Laws and Regulations": {
        "answer": "It is important to follow the
law and act responsibly when disposing " \
                    "of electrical products.
Incorrectly doing so can lead to hazards, fines, or " \
                    "in severe cases, up to two
years in prison. " \
                    "The .Gov.UK and Computer
Waste Recycling sites can provide more on this subject" ,
        "links": {
            "gov_1":
"https://www.gov.uk/guidance/regulations-waste-electrical-and-
electronic-equipment",
            "computer_waste":
"https://computerwaste.co.uk/the-complete-guide-to-weee-regula
tions-" \

"uk-for-e-waste-compliance/",

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    },
    },
    "Options of Electrical Waste Disposal": {
        "answer": "The most difficult thing about all of
this is deciding where to actually get rid of " \
        "your old devices and appliances. It is
important to do so. " \
        "to consider all of your options for the
most efficient and safest way " \
        "Whether that is going to recycling
centres, using the council collection service " \
        "or taking advantage of the Retailer
Take-Back scheme, choose what is least of a " \
        "hassle for you. For companies, the
Guidance page " \
        "of the .GOV.UK. site explains the
appropriate measures for " \
        "permitted facilities. For individuals, "
\
        "you can read the London Rubbish
Collection page to learn more.",
        "links": {
            "gov_2":
"https://www.gov.uk/guidance/waste-electrical-and-electronic"
\
"-equipment-weee-appropriate-measures-for-permitted-facilities
/" \

"4-waste-storage-segregation-and-handling-appropriate-measures
",
            "rubbish_collection":
"https://londonrubbishcollection.co.uk/blog/how-to-dispose-of-
electronics-in-the-uk/"
        },
    },
    "Health and Safety": {
        "answer": "Your safety is the most important
throughout the disposal process. Wearing PPE " \
        "(Personal Protective Equipment) and
maintaining situational awareness matters " \
        "above handling the electrical product and
transporting it. " \
        "The Health and Safety Executive page on
Waste Management and Recycling " \
        "provides an indepth collection of Health
and Safety topics to browse.",
        "gov_3": "https://www.hse.gov.uk/waste/"
    },
},

```

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    "Disposal Purposes": {
      "Are you looking to dispose of an electrical product or
battery?": {
        "Electrical product": {
          "Not damaged + functioning": {
            "answer": "If the product is in good
condition " \
                        "you can always donate it to
either a local charity " \
                        "that can accept it, CeX, if it
is a device, or even check " \
                        "if somewhere like Currys/PC
World can accept it. " \
                        "Additionally, AO can accept
large products if booked " \
                        "alongside ordering new
products, " \
                        "though delivery will cost
£30.",
            "links": {
              "currys":
"https://www.currys.co.uk/services/delivery-installation/recyc
ling.html",
              "reuse":
"https://londonreuse.org/donate/",
              "cex": "https://uk.webuy.com/sell",
              "ao":
"https://ao.com/help-and-advice/delivery-and-services"
            },
          },
          "Damaged and/or not functioning": {
            "answer": "Here you can find where you can
donate "\
                        "and dispose your large and
small products. " \
                        "Remember to check if you want
to dispose " \
                        "for Individual or Company.",
            "links_2": {
              "proskip":
"https://www.proskips.co.uk/skip-hire/south-london-depot",
              "small_product":
"https://www.bexley.gov.uk/services/rubbish-and-recycling/" \
"clothes-textiles-batteries-and-small-electricals/",
              "recycling":
"https://www.bexley.gov.uk/services/rubbish-and-recycling/",
              "tradewaste":
"https://tradewasteuk.co.uk/locations/weee-waste-removal-bexle
y/"
            },
          },
        },
      },
    },
  },

```

```

        },
    },
    "Battery": {
        "Dead and leaking": {
            "answer": "This is classified as hazardous
waste and " \
                    "will need to be disposed of
separately and safely.",
            "hazwastedisposal":
"https://www.hazwastedisposal.co.uk/hazardous-waste-disposal-1
ondon.html"
        },
        "Just dead": {
            "answer": "You will be able to find where to
dispose just dead batteries here.",
            "links": {
                "kwikfit":
"https://www.kwikfit-auto.com/blog/how-to-dispose-of-your-car-
battery",
                "halfords":
"https://www.halfords.com/motoring/advice/battery-disposal-and
-recycling.html",
                "battrecycle":
"https://www.battrecycle.org/"
            },
        },
    },
},
},

```

```

#tracking already visited questions
def menu(current, parent=None):
    while True:
        # If this level has an answer → show it
        if isinstance(current, dict) and "answer" in current:
            print("\nAnswer:")
            print(current["answer"])
            # Show links
            for key, value in current.items():
                if "http" in str(value):
                    print(f"{key}: {value}")
            input("\nPress Enter to go back...")
            return
        # Otherwise show options
        elif isinstance(current, dict):
            options = list(current.keys())
            print("\nChoose an option:")
            for i, option in enumerate(options, 1):
                print(f"{i}. {option}")
            print(f"{len(options)+1}. Go Back")

```

```

option    print(f"{len(options)+2}.  Exit")          #  Added  exit
choice = input("Enter choice: ")
try:
    choice = int(choice)
    if choice == len(options) + 1:
        # Go back to parent menu if exists
        if parent:
            return
        else:
            print("Already at the main menu.")
    elif choice == len(options) + 2:
        print("Exiting the menu. Goodbye!")
        exit()
    elif 1 <= choice <= len(options):
        selected = options[choice - 1]
        # Recursive call with current as parent
        menu(current[selected], parent=current)
    else:
        print("Invalid  choice.  Please  select  a
valid option.")
except ValueError:
    print("Please enter a number corresponding to
your choice.")
else:
    print("Something went wrong.")
    return

```

— Investigating Netflix Movies | Associate Data Scientist in Python Certification with DataCamp —

```
import pandas as pd
import matplotlib.pyplot as plt

netflix_df = pd.read_csv("netflix_data.csv")

netflix_nineties = netflix_df.loc[
    (netflix_df['type'] == 'Movie') &
    (netflix_df['release_year'] >= 1990) &
    (netflix_df['release_year'] <= 1999)
]

duration = 100

plt.hist(netflix_nineties['duration'], bins=10, color='green', edgecolor='black', alpha=0.5)
plt.title('Duration of Movies in the 1990s')
plt.xlabel('minutes')
plt.ylabel('no. of movies')

action_movies = netflix_nineties[netflix_nineties['genre'] == 'Action']

short_movie_count = 0
for label, row in action_movies.iterrows():
    if row['duration'] < 90:
        short_movie_count = short_movie_count + 1
    else:
        short_movie_count = short_movie_count
print('Short action movies make up: ' + str(short_movie_count))
```

Short action movies make up: 7

